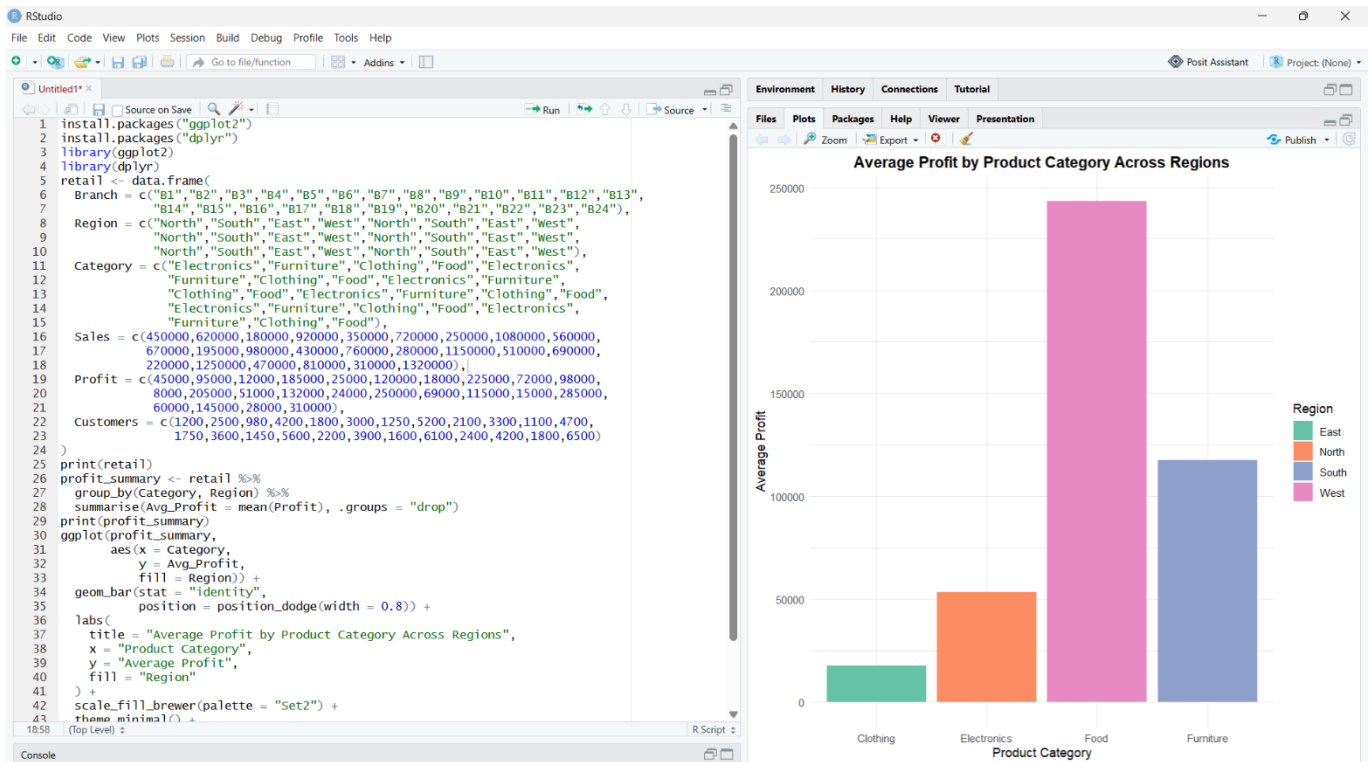


CO1_AT3_Visualization Design Exercise

Eddu Suchier-192524070

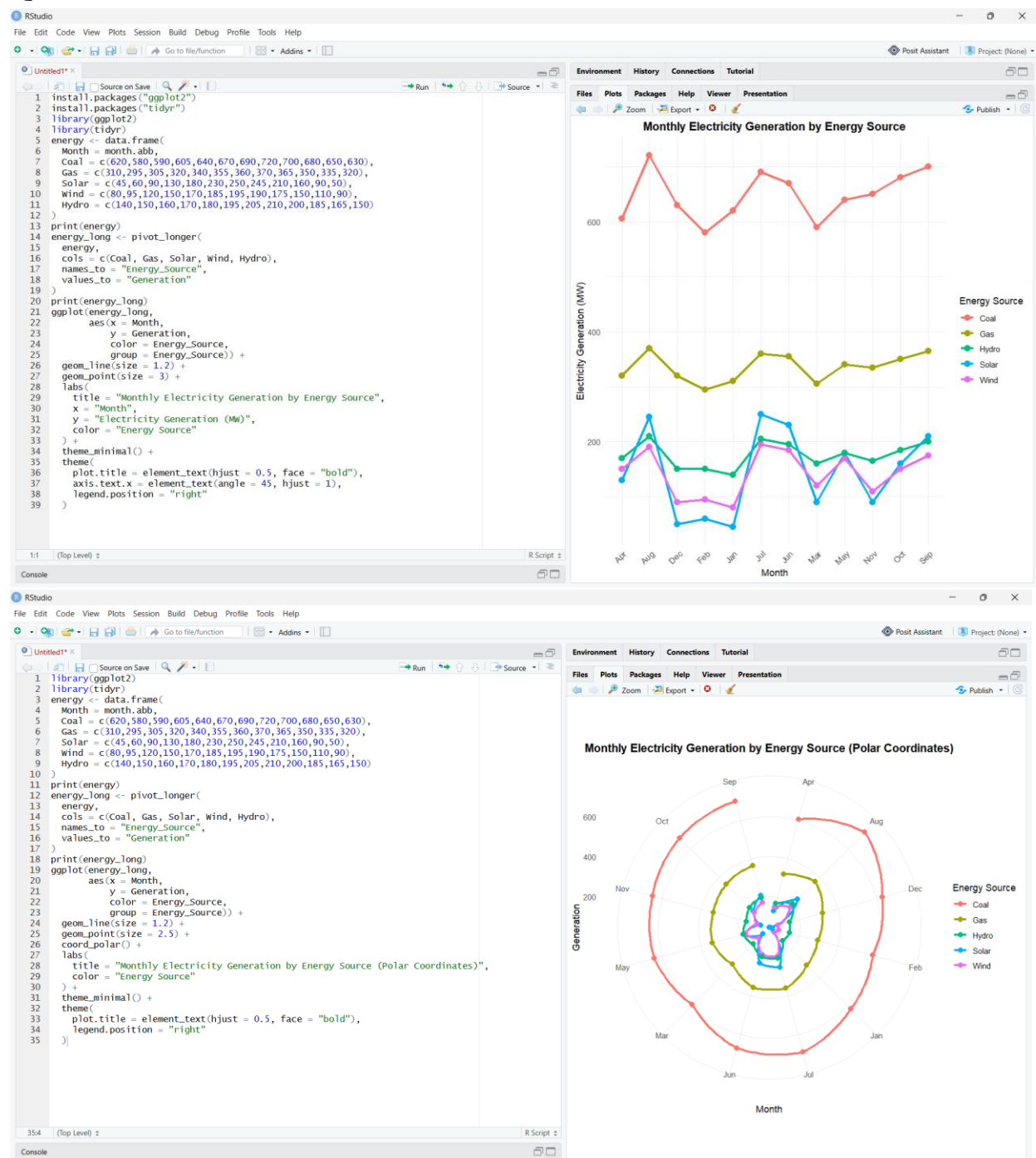
Question 1



Explanation:

1. The grouped bar chart compares the average profit across different product categories and regions.
2. Product Category is mapped to the x-axis, while Average Profit is mapped to the y-axis.
3. Different colors are used to represent each region, making comparisons easier.
4. A linear scale is used because the profit values are appropriate for direct comparison.
5. Side-by-side bars help identify which region performs better within each product category.
6. This visualization is effective for comparing multiple groups simultaneously and highlighting differences in regional performance.

Question 2

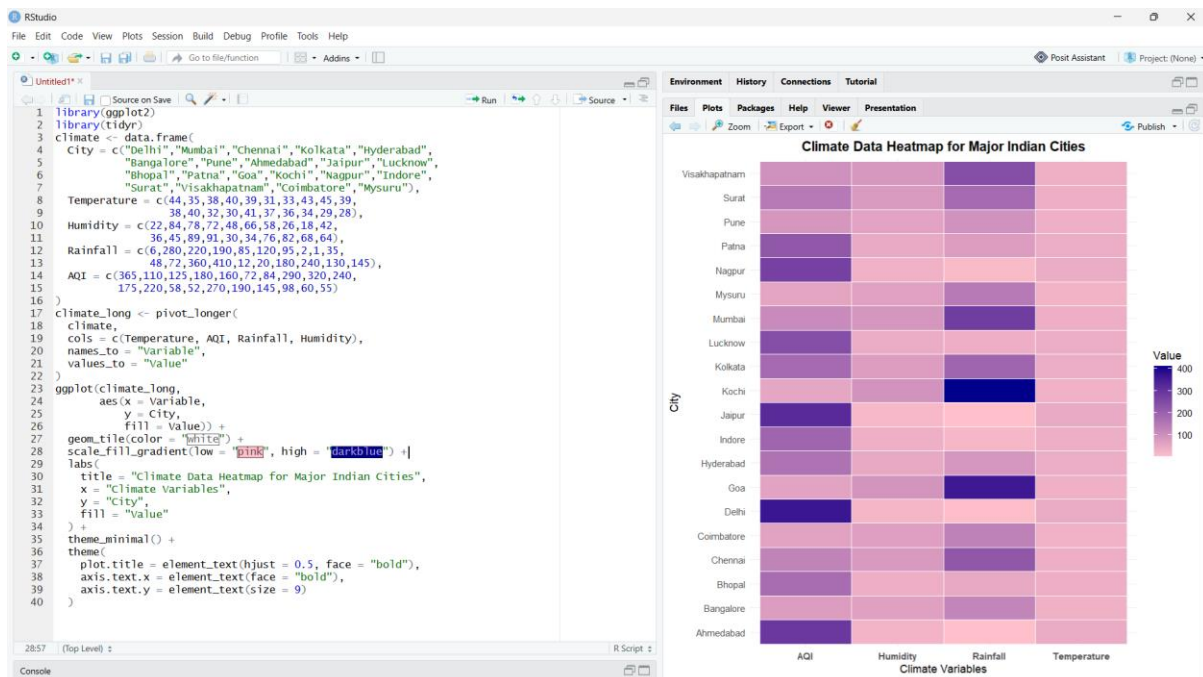


Explanation:

1. The Cartesian coordinate system clearly displays monthly trends and changes in electricity generation for each energy source.
2. The Polar coordinate system presents the same data in a circular layout, making seasonal or cyclical patterns easier to observe.
3. Different colors are used to distinguish Coal, Gas, Solar, Wind, and Hydro energy sources.

4. A linear scale is used because all generation values can be compared directly without transformation.
5. Cartesian coordinates are better for accurately comparing values and identifying trends over time.
6. Polar coordinates provide an attractive visual representation but are less effective for precise comparisons between months.

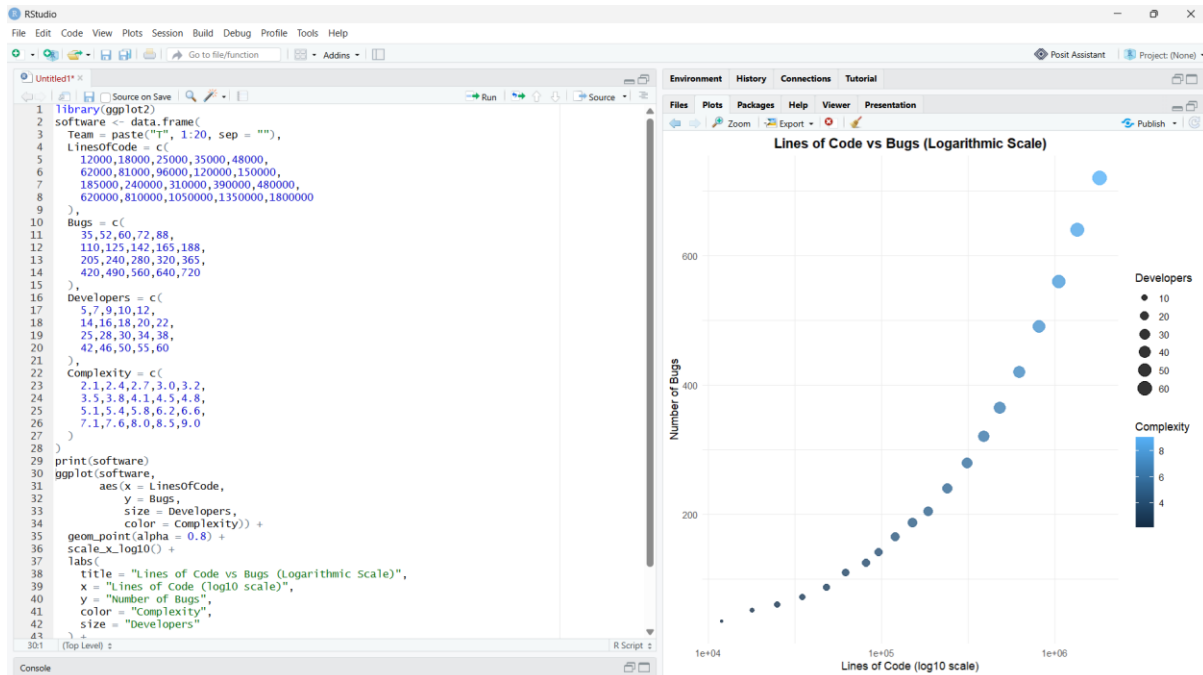
Question 3



Explanation:

1. A heatmap is used to compare Temperature, AQI, Rainfall, and Humidity across multiple cities.
2. Cities are displayed on the y-axis, while climate variables are shown on the x-axis.
3. A light blue to dark red colour gradient represents low to high values, making patterns easy to identify.
4. Darker colours indicate higher values, while lighter colours indicate lower values.
5. The colour scale improves readability and helps quickly identify cities with extreme climate conditions.
6. A heatmap is an effective visualization because it allows simultaneous comparison of multiple variables across many cities in a compact format.

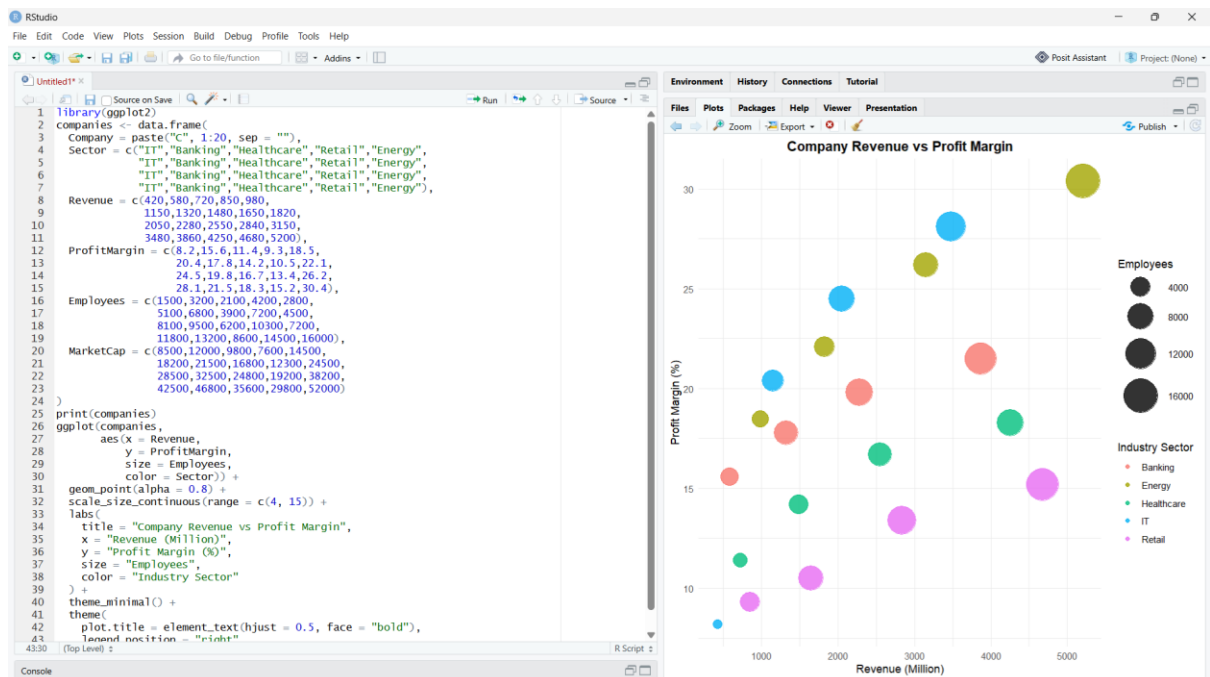
Question 4



Explanation:

1. A scatter plot is used to show the relationship between Lines of Code and Bugs.
2. A logarithmic (log10) transformation is applied to the x-axis because the Lines of Code values are highly skewed.
3. The size of the points represents the number of Developers, while the color represents Complexity.
4. The logarithmic scale compresses large values and spreads smaller values, making patterns easier to observe.
5. Compared to a linear scale, the log scale improves readability and reduces the effect of extreme values.
6. This visualization helps identify the relationship between project size, bugs, team size, and complexity more effectively.

Question 5



Explanation:

1. A bubble chart is used to visualize multiple variables in a single graph.
2. Revenue is mapped to the x-axis, while Profit Margin is mapped to the y-axis.
3. Bubble size represents Employee Strength, allowing comparison of company size.
4. Different colours distinguish the Industry Sectors, making categories easy to identify.
5. A perceptually distinct colour palette improves readability and minimizes visual bias.
6. This visualization effectively reveals relationships between revenue, profitability, employee strength, and industry sector simultaneously.